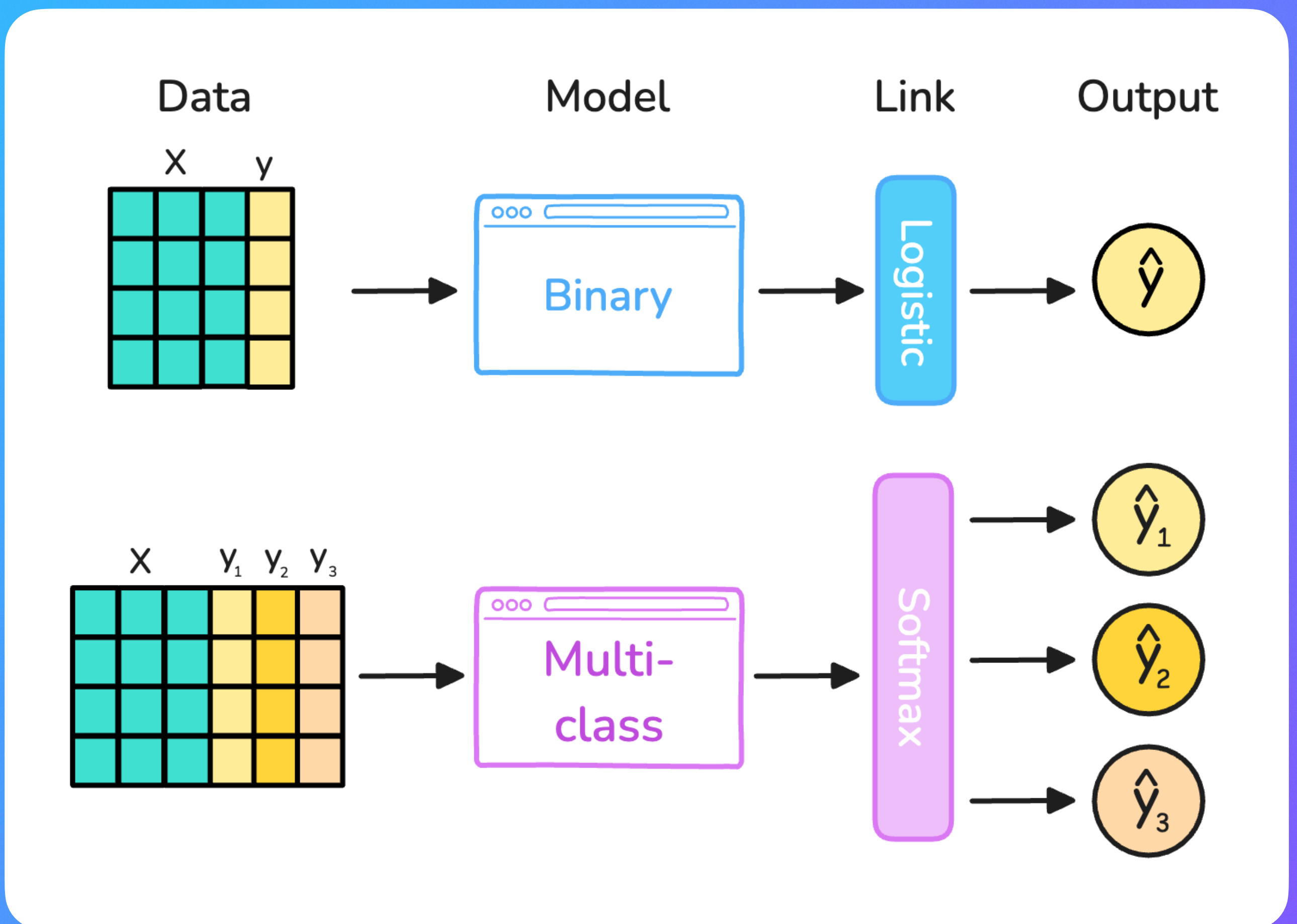




Binary and Multiclass Modeling

A lot of applied problems are modeled as a binary choice (e.g., credit decisions), which is a compact way to decompose observed real-world phenomena.

Multiclass models take this one step further by expanding the range of outcomes (e.g., the decision to approve a credit line, credit card, or a BNPL)



Binary and multiclass approaches are fundamental to modeling a wide variety of real-world scenarios because they allow us to both score and categorize outcomes, which is useful in numerous business applications

In this deck, we focus on logistic regression and gradient boosted decision tree (GBDT) models to showcase these two modeling paradigms in action ✨



Binary and Multiclass Modeling

Part 1: Linear Models

Exploring why logistic and softmax
are the same models for binary problems



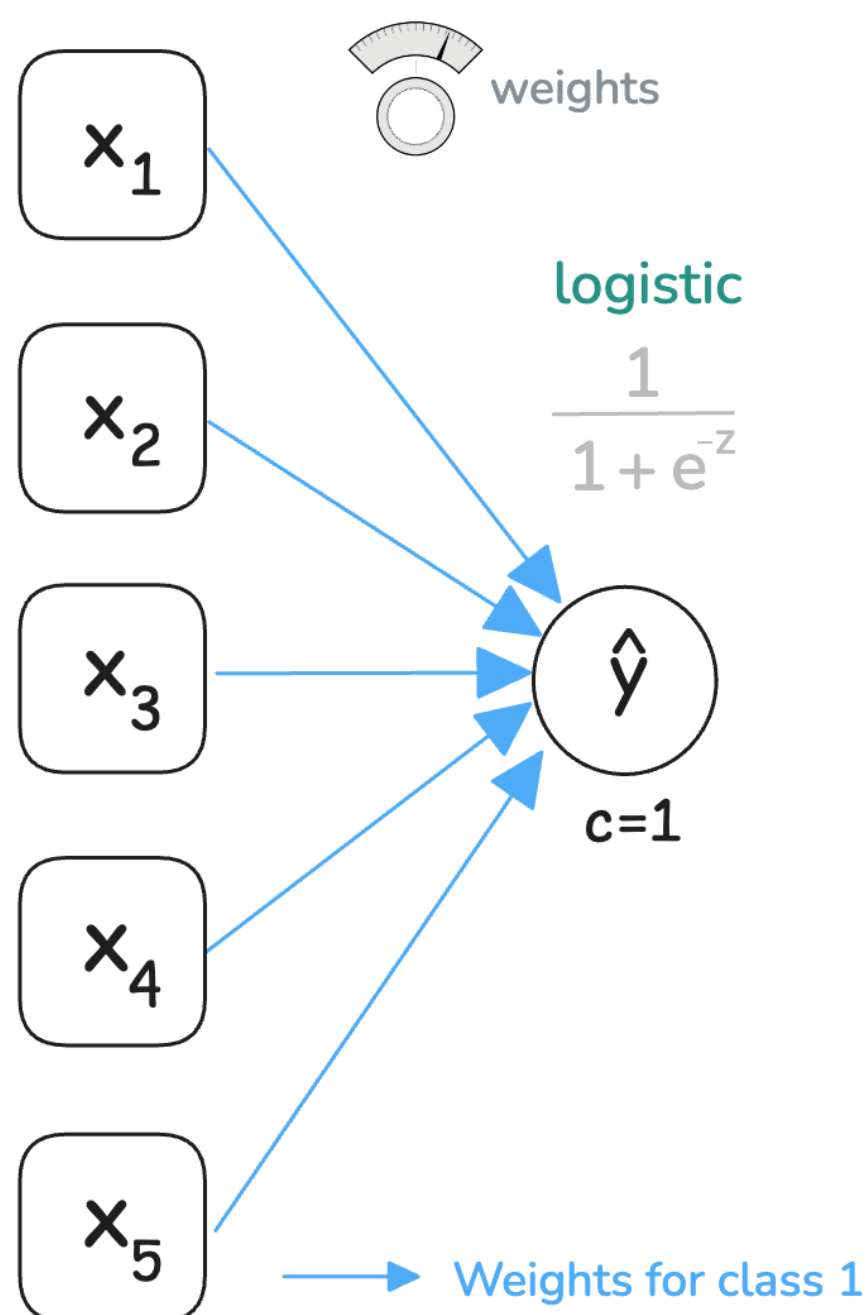
Binary and Multiclass Modeling



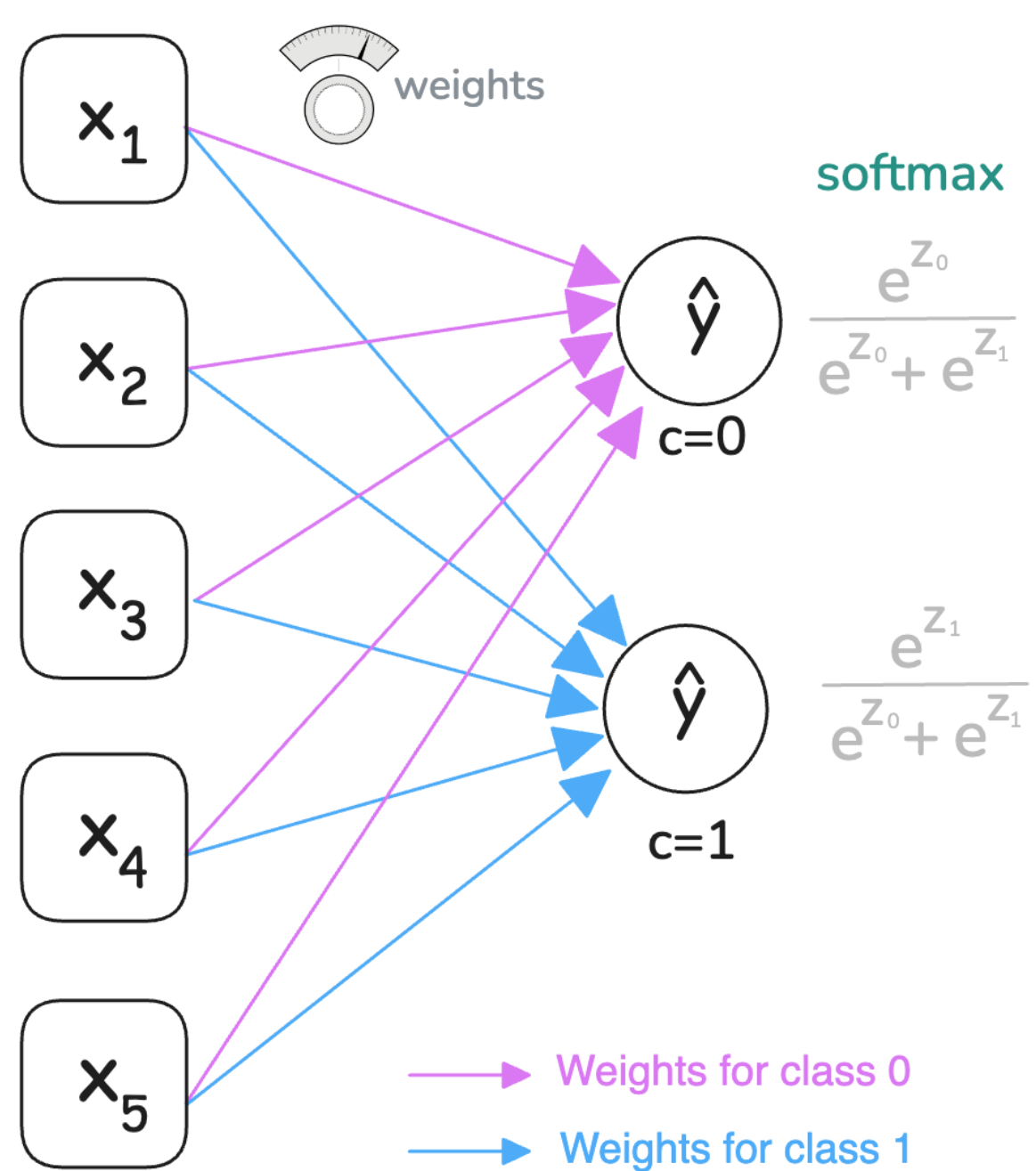
Most texts skip the important connection between binary and multiclass models for problems with two outcomes, often jumping directly to a $K > 2$ case

We can draw down these two approaches schematically, the first predicting only class 1 (binary) and the second predicting classes 0 and 1 (multiclass) ↩

Binary



Multiclass



Even though the link functions to produce class probabilities are different between the models (logistic vs softmax), they converge to the same linear decision boundary at their maximum likelihood estimate (MLE) in this case

In what follows next we introduce the equivalence of the two methods in binary problem scenarios like predicting credit default 🏦 ➡



Binary and Multiclass Modeling



We show the equivalence of the two approaches for binary problems below. For logistic regression, shown in Figure 1, we have only one column with label (1 for default or 0 for no default), p1 is class 1 probability, p0 is derived as $1 - p1$

With multinomial regression (Figure 2), we have two columns with labels for each class, and p0 and p1 are derived directly from a softmax function (not as a complement) ↘

1 Logistic regression

	sample_id	label	p0	p1
0	998	1	0.845133	0.154867
1	999	1	0.363016	0.636984
2	1000	0	0.996337	0.003663
3	1001	0	0.847788	0.152212
4	1002	0	0.988499	0.011501

2 Multinomial regression

	sample_id	label=0	label=1	p0	p1
0	998	0	1	0.845133	0.154867
1	999	0	1	0.363016	0.636984
2	1000	1	0	0.996337	0.003663
3	1001	1	0	0.847788	0.152212
4	1002	1	0	0.988499	0.011501

For consistency in MLE estimates, in this deck we use the fisher-scoring package that produces coefficients and standard errors for both types of models¹

1. <https://pypi.org/project/fisher-scoring/>



Binary and Multiclass Modeling

If we inspect the weights (coefficients) of the two models (logistic and multinomial regression) for binary classification closer, we discover that a) both models produce identical Gini scores and b) the weights of logistic regression are twice the weights of multinomial regression for class 1 ↷

Logistic regression

Weights for class_1

Feature	Weight
x1	5.1540
x2	2.8504
x3	-0.0969
x4	0.0238
x5	0.0376
x6	-0.0154
x7	0.0000

Gini score: 79.32%

Multinomial regression

Weights for class_0

Feature	Weight
x1	-2.5770
x2	-1.4252
x3	0.0485
x4	-0.0119
x5	-0.0188
x6	0.0077
x7	-0.0000

Weights for class_1

Feature	Weight
x1	2.5770
x2	1.4252
x3	-0.0485
x4	0.0119
x5	0.0188
x6	-0.0077
x7	0.0000

Gini score: 79.32%

Logistic class 1 weights:

Weight 1: 5.1540
Weight 2: 2.8504
Weight 3: -0.0969
Weight 4: 0.0238
Weight 5: 0.0376
Weight 6: -0.0154
Weight 7: 0.0000

equivalence
of
weights



Multinomial class 1 weights * 2:

Weight 1: 5.1540
Weight 2: 2.8504
Weight 3: -0.0969
Weight 4: 0.0238
Weight 5: 0.0376
Weight 6: -0.0154
Weight 7: 0.0000

From the outputs above we see that in order to transform multinomial regression into a logistic regression in a binary setting we need to multiply the weights and biases (coefficients and intercepts) by the factor of 2¹

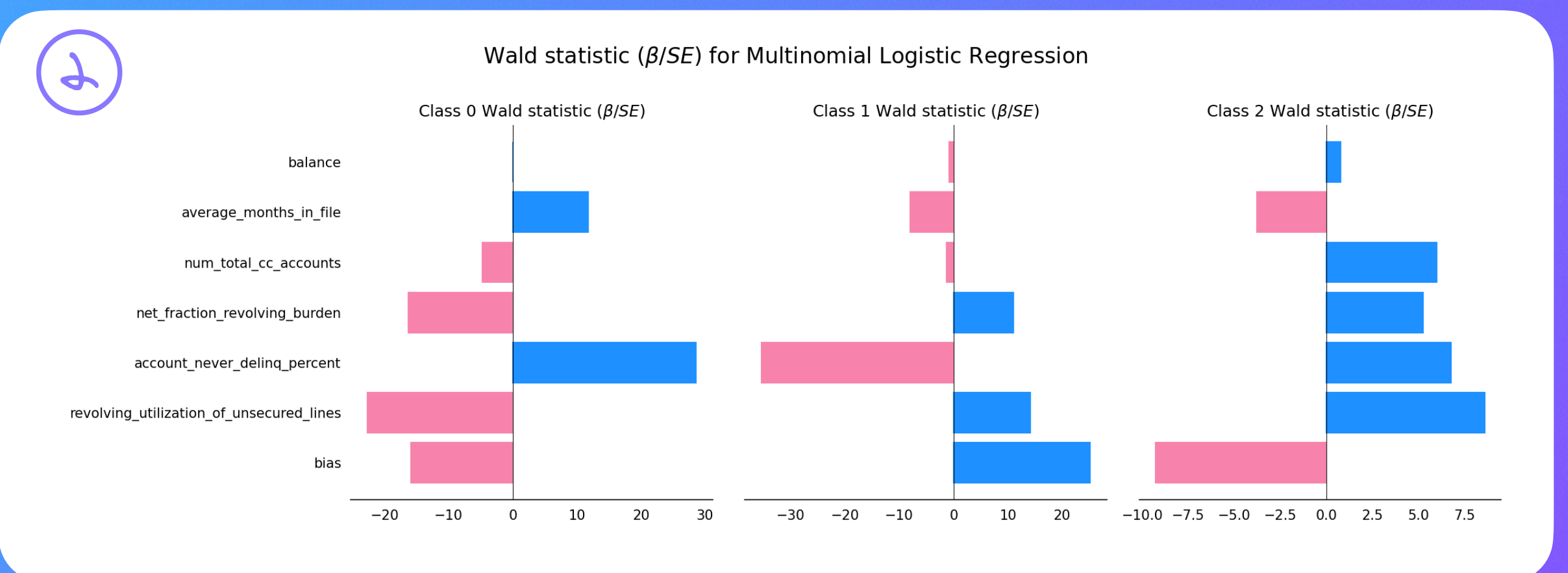
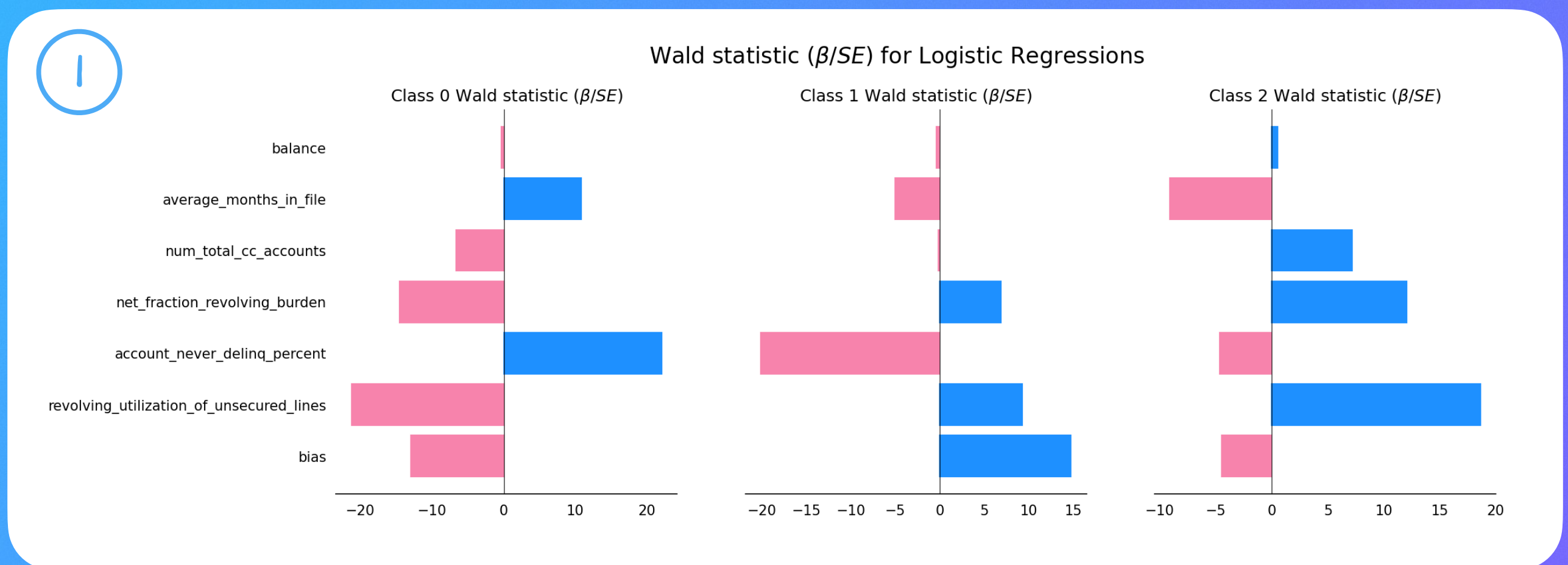
1. You can find more details about this transformation in this deck: <https://tr.ee/ptDQ4hqqSZ>



Binary and Multiclass Modeling

We can extend our two-class credit risk problem to three classes, for instance, if we split our binary default flag into days past due (DPD) and unlikely to pay (UTP) default reasons as per standard regulatory treatment

Below we show the importance of features for the two modeling approaches: one vs rest logistic regressions (1) and multinomial regression (2) ↘



Wald scores (coefficients divided by standard errors) are similar for class 0 (no default) and class 1 (UTP default) between models, while for class 2 (DPD default) they differ significantly due to the multinomial model's limited expressivity ➡



Binary and Multiclass Modeling

Part 2: Gradient Boosting 🌳

Turning multiclass GBDT models
into high-performing binary classifiers



Binary and Multiclass Modeling



The issue with linear models for classification is that due to the linear nature of weights (coefficients), the model loses a lot of flexibility in modeling classes

We can clearly see this if we compare performance of logistic regression with the multinomial logistic regression on the same underlying data ↘

Gini and F1 Scores for Logistic Regression

Probability	Gini Score	F1 Score
prob_0	0.79	0.95
prob_1	0.79	0.40

Gini and F1 Scores for Multinomial Regression

Probability	Gini Score	F1 Score
prob_0	0.77	0.95
prob_1	0.95	0.45
prob_2	0.71	0.14
prob_1_and_2	0.77	0.39

Notice the drop in rank-ordering performance (Gini) for multinomial logistic regression model vs logistic regression due to the inclusion of more classes

The prob_1_and_2 is the complement of prob_0 (class 0 prediction), because due to the linear properties of multinomial regression, the weights with a reversed sign give the probability of all other classes in a multinomial model¹

1. For a good discussion on GLM see: <https://forum.posit.co/t/understanding-glmnets-multinomial-logistic-regression-coefficients-not-k-1/52031>



Binary and Multiclass Modeling



When using linear models, turning a binary problem into a multiclass one reduces performance for the defaulted vs non-defaulted class. However, more flexible ML algorithms, like gradient boosted decision trees (GBDT), can leverage class information to improve binary ranking performance

We repeat the same experiment with XGBoost and CatBoost using the same set of hyperparameters for comparison, i.e. 100 decision trees with a depth of 2

Gini and F1 Scores for XGBoost (binary)

Probability	Gini Score	F1 Score
prob_0	0.88	0.96
prob_1	0.88	0.53

Gini and F1 Scores for XGBoost (multiclass)

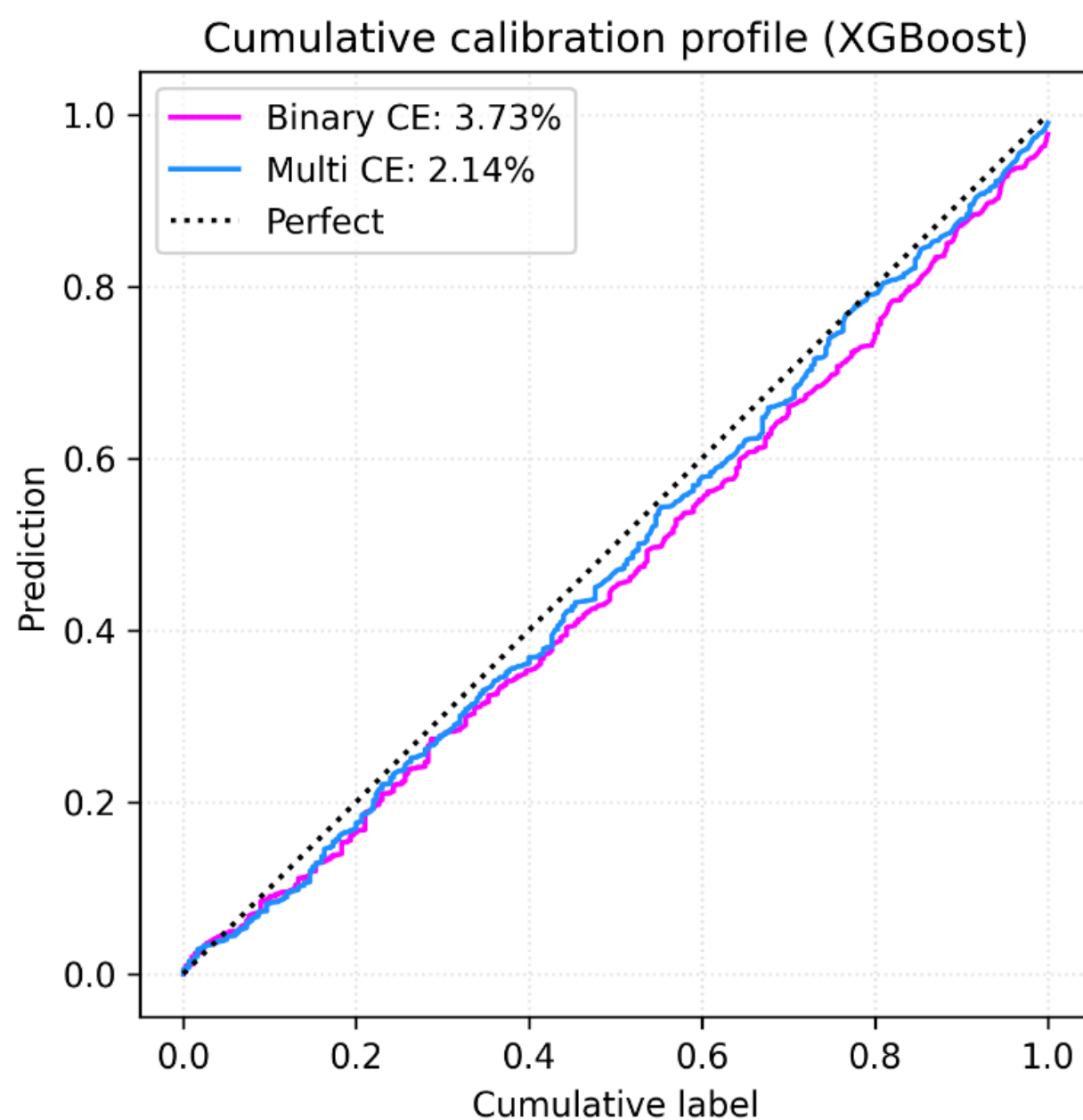
Probability	Gini Score	F1 Score
prob_0	0.88	0.96
prob_1	0.98	0.66
prob_2	0.90	0.51
prob_1_and_2	0.88	0.56

It turns out that using a multiclass GBDT model can actually give a noticeable improvement in terms of the discriminatory power over default vs non-default risk differentiation

Next, we'll see how calibration accuracy also improves with this approach ➡



Binary and Multiclass Modeling



Cumulative calibration profile

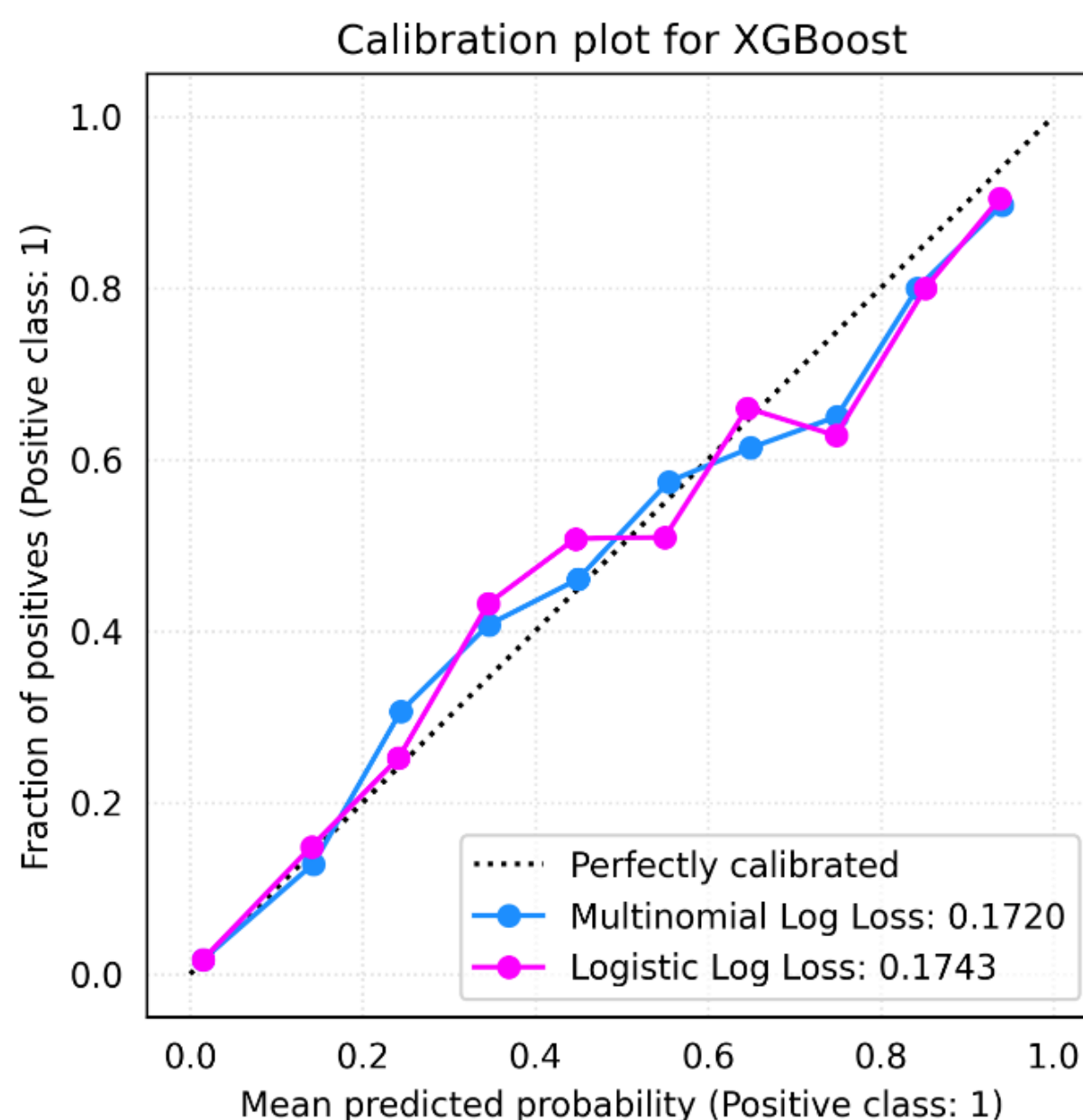
This plot shows a continuous view of the calibration of raw outputs from a binary XGBoost model and a multiclass one (we sum probabilities of class 1 and 2)

Notice that calibration error (CE) is better for multiclass confirmed by the line closer to the diagonal

Binned calibration plot

To the right we show a binned version of a calibration plot as well as log loss results for the two models

We can confirm that the multiclass model delivers better log loss scores while the plot itself gives some mixed results due to different probability distributions





Binary and Multiclass Modeling

Below we show the results for CatBoost, another GBDT famous library, which differs from XGBoost in several ways, but mainly when it comes to processing categorical data

For practitioners, a choice between CatBoost and XGBoost must be driven by dataset characteristics and complexity, but both algorithms in general should yield the best performance on the tabular datasets among other options

Gini and F1 Scores for CatBoost (binary)

Probability	Gini Score	F1 Score
prob_0	0.88	0.96
prob_1	0.88	0.51

Gini and F1 Scores for CatBoost (multiclass)

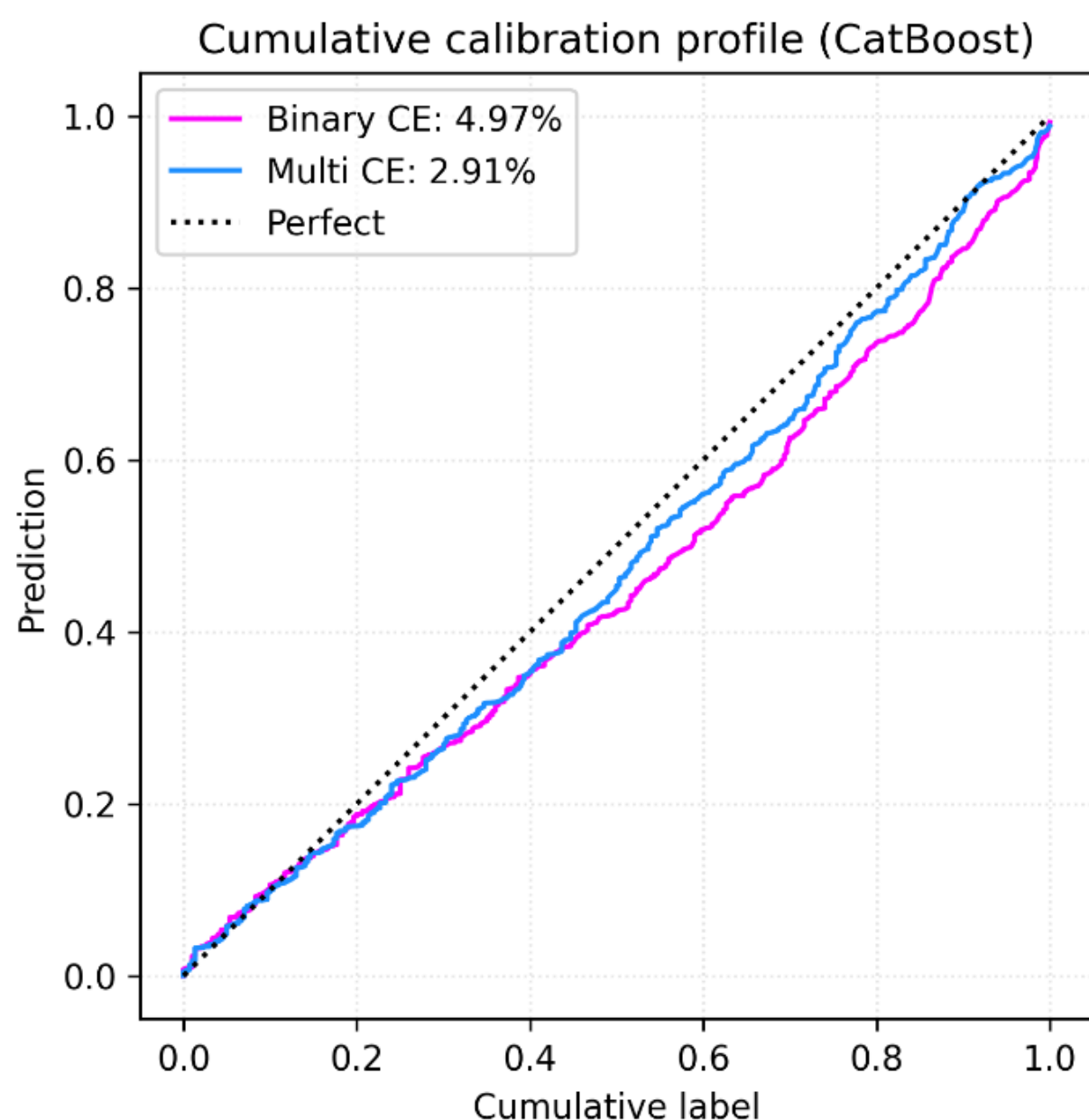
Probability	Gini Score	F1 Score
prob_0	0.88	0.96
prob_1	0.98	0.65
prob_2	0.90	0.51
prob_1_and_2	0.88	0.55

Here we see somewhat similar results as with XGBoost in that splitting the defaulted data into two more classes allows CatBoost to increase the performance of the binary ranking

As before, we'll analyze the differences in calibration accuracy for CatBoost ➡



Binary and Multiclass Modeling



Cumulative calibration profile

This plot shows a continuous view of the calibration of raw outputs from a binary CatBoost model and a multiclass one (we sum probabilities of class 1 and 2)

Again, notice that calibration error (CE) multiclass is best and more aligned with the diagonal

Binned calibration plot

To the right we show a binned version of a calibration plot as well as log loss results for the two models

The multiclass CatBoost model appears to be of great use for binary classification meaning that it delivers more accurate predictions than just binary one

